

RESEARCH METHODOLOGY

SAMPLING TECHNIQUES AND
DATA COLLECTION METHODS

Presented by **DR TUMPA BANERJEE**



www.sittechno.org

22-09-2026

by Dr. Tumpa Banerjee



Census vs. Sample Survey

Core Concept: How do we collect data from a universe of interest?

Universe / Population (N): The total collection of all items/elements in a given field of inquiry.

Census Survey: A complete enumeration of every single item in the population.

- *CS Example:* Analyzing every single execution log row in a multi-terabyte system log or scanning every line of code in an enterprise repository for security flaws.
- *Drawbacks:* High cost, immense computational/time requirements. If systematic bias exists, it grows larger as observations increase.

Census vs. Sample Survey

Sample Survey ($n < N$): Selecting a smaller representative subset (n) from the population (N) to make inferences.

- *CS Example:* Profiling 5% of API requests or running automated tests on a selected set of module functions.
- *When to use:* Necessary when the population is large/infinite or resources (compute, time, budget) are constrained.

Principles of Sample Design

A sample design is a definite plan for obtaining a representative sample from a given population before any data collection begins.

Steps in Developing a Sample Design:

- **Type of Universe:** Define whether the population is **Finite** (e.g., registered database users) or **Infinite** (e.g., continuous stream of IoT sensor packets).
- **Sampling Unit:** The base element being selected (e.g., individual user, server node, database record).
- **Sampling Frame (Source List):** The complete list/directory from which the sample is drawn (e.g., table of user IDs).
- **Size of Sample (n):** Balancing precision, confidence level, population variance, and computational/budgetary constraints.
- **Sampling Procedure:** Choosing the exact algorithm or method to draw elements.

Errors in Sampling & Good Design Criteria

- **Systematic Bias:** Results from flaws in sampling frames, defective measuring instruments/questionnaires, non-responses, or natural reporting bias. *Cannot be eliminated by increasing sample size.*
- **Sampling Error:** Random variations of sample estimates around true population parameters. *Decreases as sample size (n) increases.*
- **Characteristics of a Good Sample Design:** High representativeness, minimal sampling error, cost/viability balance, controlled systematic bias, and generalizable results.

Taxonomy of Sampling Methods

Selection Technique	Probability Sampling (Random Selection)	Non-Probability Sampling (Non-Random)
Unrestricted Sampling	Simple Random Sampling	Convenience / Haphazard Sampling
Restricted Sampling	Stratified, Systematic, Cluster, Multi-Stage	Purposive, Quota, Judgment Sampling

Probability Sampling Methods

Definition: Every item in the population has a known, equal chance of inclusion ($1/N$) via a mechanical/chance process.

- **Simple Random Sampling:** Each element and sample combination has an equal probability of selection ($1/\binom{N}{n}$).
 - *CS Technique:* Drawing pseudo-random numbers (PRNG) or using Tippett's random number tables to sample database IDs.
- **Systematic Sampling:** Selecting every k^{th} item from an ordered list ($k = N/n$) after a random start.
 - *CS Example:* Sampling every 50th incoming packet in network traffic analysis.
 - *Risk:* Hidden periodicity in data can introduce severe bias.

Advanced Probability Sampling Methods

Stratified Random Sampling: Population is divided into non-overlapping homogeneous sub-groups (**strata**), and random samples are drawn from each.

- **Proportional Allocation:** Strata sample sizes are proportional to sub-population sizes ($n_i = n \times \frac{N_i}{N}$).
- **Optimum Allocation (Neyman):** Accounts for varying standard deviations (σ_i) across strata:
 - $$n_i = \frac{n \cdot N_i \sigma_i}{\sum N_j \sigma_j}$$

Advanced Probability Sampling Methods

- **Cluster & Area Sampling:** Divide population into heterogeneous groups (**clusters**); randomly select whole clusters to survey.
 - *CS Example:* Sampling 5 server racks (clusters) out of 100 in a data center and auditing all VMs within them.
- **Multi-Stage & Sequential Sampling:** Sampling in multiple steps (e.g., Data Center → Server → VM) or sampling continuously until a decision threshold is met.

Non-Probability Sampling Methods

Definition: Selection relies on researcher discretion or availability rather than random chance; probability of selection cannot be determined mathematically.

- **Convenience / Haphazard Sampling:** Choosing items based on ease of access.
 - *CS Example:* Testing a newly developed UI only on your classmates' smartphones.
- **Purposive / Judgment Sampling:** Deliberately choosing items deemed typical or relevant to the study.
 - *CS Example:* Benchmarking database query speed on specific edge-case servers known to experience high bottlenecks.

Non-Probability Sampling Methods

- **Quota Sampling:** Assigning target quotas to interviewers to fill from specific demographics at their discretion.
- **Key Limitation:** Inferences cannot be formally evaluated for sampling error or statistical significance.

Data Collection: Primary vs. Secondary Data

- **Primary Data:** Original data collected for the first time by the researcher directly from source (e.g., custom benchmark runs, fresh user interviews, survey responses).
- **Secondary Data:** Data already collected, compiled, and published by others (e.g., public GitHub repositories, Kaggle datasets, system event logs, published literature).

Method of Primary Data Collection

Primary data may be collected through

- Observation
- interviews
- Questionnaires
- Schedules

Observation Methods

Observation methods is a method under which data collected with the help of observation by the observer.

Type of observation method

- Structured
- Unstructured
- Participant
- Non participant

Structured Observation

1. Structured Observation

- Systematic observation
- For structured observation the researcher specifies in details what is observed and how the measurement are to be recorded.

2. unstructured observation

- observe and record behavior in a holistic way without the use of predetermined guide.

3. Participant observation

- when the observer is a member of the group which he is observing then it is participant observation

4. Non participant observation

- When the observer is not a member of the group
- Observer is observing people without giving any information to them then it is non participant observation.

Advantage and Disadvantages of Observation Method

Advantages

- Produces large quantities of data
- All data obtained from observations are usable
- The observation technique can be stopped or begin at any time
- Relatively inexpensive

Disadvantages

- Time consuming
- Limited information
- Extensive training is needed
- Not use for studying past event or activities

Interview Method

Interview is a process of communication or interaction between two person to collect the relative information for the research.



Type of interview method

- Personal interview, it can be
 - Structure
 - Unstructured
- Telephonic interview

Advantages

- More information at greater depth can be obtained
- The interviewer can judge the non-verbal behavior of the respondent
- The interviewer can control over the order of the question as in the questionnaire and can change the spontaneity of the respondent as well
- Personal information can be obtained

Disadvantages

- It is an expensive method
- Interviewer buyers
- Respondent bias
- Time consuming
- There is a lack of accessibility to respondents

Questionnaires

- Questionnaire is as an instrument for research
- It consists of a list of questions, along with the choice of answers, printed or typed in a sequence on a form used for acquiring specific information from the respondents
- Questionnaire are delivered to the concerned person either by post or mail, requesting them to answer the questions and return it.

Advantage of questionnaire

- Questionnaires lies in their uniformity i.e. is all respondents see exactly the same questions
- It is an expensive method
- Free from the bias of the interviewer, as the respondent answers the questions in his own words
- Respondents have enough time to think and answer
- Due to its large coverage, respondents living in distant areas can also be reached conveniently
- Large samples can be used.

Disadvantage of questionnaire

- Time consuming
- The respondents need to be educated and cooperative
- Possibility of unclear replies
- The rate of non-response is high

Schedules

- Schedule is the tool or instrument used to collect data from the respondent while interview is conducted
- In schedule there is an enumerator who filled the questionnaire
- Enumerator goes to the respondent, asks them the questions from the questionnaire in the order listed and records the response in the space provided
- Enumerator must be trained in administering the schedule

Difference between questionnaire and schedule

Questionnaire	Schedule
it is generally sent through mail and no further assistance from the sender	Schedule is filled by the enumerator or research worker
It is cheaper method	Costly requires field worker
Non-response is high	Non-response is low
In questionnaire, it is not confirmed that expected respondent have filled the answers	In schedule identity of person is known

Designing Questionnaires & Schedules

Designing reliable survey instruments requires structured preparation to avoid systematic response bias.

- **Key Rules for Effective Design:**

- **Question Structure:** Keep questions simple, clear, non-ambiguous, and direct. Avoid leading or double-barreled questions.

- **Question Types:**

- *Closed-Ended:* Multiple choice, rating scales (Likert scale), binary (Yes/No)—ideal for quantitative metrics.
- *Open-Ended:* Text boxes—ideal for qualitative feedback on system usability.

Designing Questionnaires & Schedules

- **Logical Flow:** Start with broad introductory questions, move to specific technical queries, and place sensitive questions at the end.
- **Pre-Testing / Pilot Study:** Run a small preliminary test ($n = 10 - 20$) to identify confusing questions and measure completion time before full deployment.

Secondary method of data collection

- data gathered and recorded by someone else
- It involves less cost, time and effort
- Secondary data is data that is being reused. Usually in different context
- For example: data from a book, magazines, and newspapers
- It can be collected from internal and external sources

Internal sources

- Sales records
- Marketing activity
- Cost information
- Distributor reports and feedback
- Customer feedback

External source

- Books and magazines journals
- Newspaper
- Libraries
- The Internet or website
- Publish data by government or private organization
- Census etc..

Factors to be considered before using secondary data

- Reliability of data- who, when, which method, at what time etc. must be investigate
- Suitability of data- object, scoop, and nature of original inquiry need to be studied, as if the study was with different objective then the data is not suitable for current study
- Adequacy of data – label of accuracy, area differences then data is not adequate for study

Advantage of Secondary data collection

- Ease of access
- low cost or free
- Time saving
- Allow you to generate new insights from previous analysis
- Longitude then denial analysis
- A huge amount of secondary data with a wide variety of sources

Disadvantage of secondary data

- Might be not specific to your needs
- You have no control over data quality
- Bias
- Not timely

Selection of proper method for collection of data

- Nature, scope and object of inquiry
- Availability of funds
- Time factor
- Accuracy required

Summary

- **Sampling Summary:** Use probability sampling (Random/Stratified) for statistical rigor and sampling error control. Use non-probability sampling for quick feasibility studies.
- **Data Collection Summary:** Ensure survey tools and measurement code are pre-tested to prevent systematic bias.

THANK YOU

WE APPRECIATE YOUR TIME AND
ATTENTION TO OUR RESEARCH STUDY.

Presented by **DR TUMPA BANERJEE**

+91 9883057767

www.sittechno.edu.in

Tmp.manna@gmail.com

22-09-2026

by Dr. Tumpa Banerjee

